The Sound Shell

Team MALINE
Mitul Pandya
Ayan Kadri
Elizabeth “Lizzie” Hoole
Ivan Li
Nicholas “Cole” Miller
Eric Son

Section ME 4182 – A
Dr. Amit Jariwala

Advisor: Dr. David MacNair
Georgia Institute of Technology
North Ave NW, Atlanta, GA 30332

Captain Knapp, Captain Cooper (National Security Innovation Network NSIN)
Moody Air Force Base Bemiss Rd, Moody AFB, GA 31699
ryan.knapp.4@us.af.mil, jonathan.cooper.14@us.af.mil
Captain Cooper Phone Number: +1 (864) 569-4043

Table of Contents

Executive Summary ii
Nomenclature iii
Main Body 1
1) **Introduction and Background** 1
2) **Existing Products, Prior Art, and Applicable Patents** 2
3) **Codes and standards** 3
4) **Customer Requirements and Engineering Specifications** 5
5) **Market Research** 8
6) **Proposed Design Solution** 8
7) **Industrial Design** 12
8) **Testing** 13
9) **Material Selection** 18
10) **Manufacturing** 20
11) **Assembly** 21
12) **Future Considerations** 22
13) **Conclusions, Future Work, Project Deliverables** 24
14) **Team Member Contributions** 25
15) **Acknowledgements** 27
Appendix 28
References 30

Executive Summary

The helicopter rescue missions carried out by the Air Force are fast-paced situations that require equipment that is reliable, effective, and easy to use. Much of this equipment's intended use is to ensure the rescued patients, who may be suffering from critical injuries, are as safe and comfortable as possible. One potential hazard these patients face during a rescue is hearing loss, as the environment in and around a helicopter can be dangerously loud, and standard headphones may not be compatible with the type of injury the patient is sustaining. The goal of the project described in this report is to work towards a product that solves this by providing hearing protection for injured patients, specifically those being transported by the 41st Rescue Squadron of the U.S Air Force.

The design team, MALINE, has selected to pursue a shell-like device that can attach to a Stokes Litter and comfortably fit around the patient's head without enclosing it tightly. The design will feature sections that can be easily opened and closed if access to the patient's head is necessary, alongside semi-transparent walls so medics can maintain constant visual on the patient. The ability of this solution to accomplish the client's goal is based on existing technology for passive noise cancellation, with the potential to include active noise cancellation for future enhancements. One specific goal of this device is to achieve a 10 dB reduction in noise. The means of accomplishing this is determined by the materials chosen to constitute the device. Through research and testing, the chosen materials are Bass Buster acoustic insulation foam and cost-effective cured silicone plastic. Additional goals of safety and comfort were met through geometrical design features that allow for complete enclosure of the patient's head.

Nomenclature

- **Acoustic Impedance**
  - The ratio of the acoustic pressure and velocity of a wave. Specific to material type, and higher differences in acoustic impedances between.
- **Active Noise Control (ANC)**
  - This is a method of reducing a certain sound by adding a second source of sound with a negative amplitude of the initial one to create a dampened residual noise.
- **dBA**
  - Refers to decibel measurements with the A-weighting scale. A-weighting is based upon the frequency sensitivity of the average human ear and weights high-frequency noises more than low-frequency noises.
- **dBC**
  - Refers to decibel measurements with the C-weighting scale. C-weighting is based upon the frequency sensitivity of the average human ear at high noise levels, weighing all frequencies at a similar level.
- **Noise Reduction Coefficient (NRC)**
  - A measure of how well a material absorbs sound. This value ranges from 0 to 1, with 0 meaning no sound is absorbed and 1 meaning all sound is absorbed [1].
- **Passive Noise Control (PNC)**
  - This is a method of reducing general sound by letting the sound wave pass through an absorber to create a dampened residual noise.
- **Stokes Litter**
  - This is a rescue basket that is designed to carry and strap on injured, sick, or disabled people.

Main Body

1) Introduction and Background

The sound produced by military helicopters often exceeds dangerous levels loud enough to warrant all crew members wearing advanced over ear headphones. Injured medivac patients, however, may not be provided these headphones based on limited time or conflicting head injuries. The design problem for this project is to create a device or environment that can provide hearing protection for the patients that the 41st Rescue Squadron of the U.S Air Force rescues with helicopters. During these rescue missions, sound pressure levels experienced by the patient can exceed 130 dBC due to noise created by the rotor, engine, and auxiliary power unit of the helicopter [2]. MALINE's product will provide a solution that reduces the noise perceived by the patient. Additionally, the device will be able to accommodate a variety of patients, be implemented quickly, and permit medics to conduct their necessary procedures.

The solution MALINE is pursuing is a shell-like device that relies primarily on passive noise control by using several acoustic-damping material layers. It is designed to fit onto the Stokes litter before the patient is placed on, and specific importance has been given to ensuring it is light, transportable, and spatially conservative. Specific details of the design are to be discussed in later sections of the report, but key features include a semitransparent shell that provides both noise reduction and visibility, support and securement for the head and neck, and compatibility to various environments including seawater.

The primary technical issue in developing this product is providing hearing protection while simultaneously interfering with the patient's body and environment as little as possible. There are numerous hearing protective solutions that currently exist such as over-ear or in-ear headphones, but these would not be compatible with all patients who have suffered head or neck injuries where their ears need to remain unobstructed. An additional challenge is creating a device that is portable and cost-effective. With an estimated weight of 55.2 lbs. (which is known to be an overestimate for conservativeness), a volume of 2047.66 in^3 (roughly the volume of a small microwave), and a clearance from the average patient's head of 3.56 inches, MALINE's device satisfies these primary design challenges using light yet durable plastics and acoustic foams and a strategically made design.

The successful creation and implementation of this device would most directly benefit the injured survivors being rescued by the rescue team. Hearing damage that results from the extremely loud conditions during their rescue would be one less potential injury. Additionally, the Air Force crew members and medics carrying out the rescue would have one less health-related issue to be concerned with during the mission. Beyond the scope of this project, this device also has the potential to be employed by other organizations who work with caring for injured patients where hearing loss may be an issue, such as non-military helicopter rescue groups.

This report will outline existing technology relating to this problem, codes, and standards the device must comply with, and requirements and specifications the product must meet. Additionally, further details on the current design and methods of testing its viability both in meeting technical specifications and the customer's requirements/desires will be described. Finally, plans for future work and further additions that can be made to the device will also be provided.

2) Existing Products, Prior Art, and Applicable Patents

There is currently a wide range of technology that exists for the purpose of reducing perceived noise levels. Although no devices currently exist that solve the design challenge for this project, inspiration was drawn from several of them in the development of the selected design. Current headsets on board the search and rescue helicopters that the Moody Air Force Base commonly use are the David and Clark headphones. These headphones use active noise cancellation methods, but primarily passive noise via specially designed ear covers that can cover the ear and dissipate sound very well. The David Clark Company lists their H10-76 headsets, shown below in Figure 1, as having Noise Reduction Ratings (NRR) of 24 dB [3]. Several other examples of over and in-ear headphones such as those created by Bose and Apple were also researched to understand the materials and methods used. A limitation with these devices is that they completely obstruct the ear as well as the area around it, which makes them unusable for patients with head or neck injuries. However, the use of the acoustic foam shaped to a specific geometry and sealed in rubber or plastic, which is often used in over-ear headphones, was a significant feature the team plans to utilize in the selected design.



Figure 1. David and Clark's H10-76 headset commonly used by the crew in search and rescue operations for the Moody Air Force Base.

Some inspiration for initial ideation also evolved from products seen on popular markets such as Amazon. For example, the NoCry 6-in-1 safety helmet, as shown in Figure 2, represents the backbone of

MALINE's initial design ideation. The product is comprised of a helmet and transparent face shield which allows for multiple areas of attachment points and for the device to sit comfortably on the head of the user. As seen in the photo below, the headphones can be easily attached to the helmet [4]. The selected design MALINE is moving forward with is a more advanced version of such a helmet that provides a slightly more robust volume to encapsulate the head so that there are fewer openings for sound transmission and headphone pads do not need to be in direct contact with the patient's head.



Figure 2. NoCry 6-in-1 safety helmet which features a helmet, headphones, and a face shield [4].

3) Codes and standards

For this device, relevant codes and standards pertain to its ability to provide adequate hearing protection as determined by Occupational Safety and Health Administration (OSHA) requirements. Through communications with the Moody Air Force base, the hearing protection used by helicopter personnel follow these requirements. OSHA standards state that hearing protection must be provided to employees exposed to working environments that exceed a specific decibel threshold that is exposure time dependent. These standards and thresholds are shown below in Table 1 provided by the OSHA 1910.95 standard [5].

Table 1. Official OSHA 1910.95 standard for hearing protection [5].

TABLE G-16 - PERMISSIBLE NOISE EXPOSURES (1)

Duration per day, hours	Sound level dBA slow response
8	90
6	92
4	95
3	97
2	100
1 1/2	102
1	105
1/2	110
1/4 or less	115

Footnote(1) When the daily noise exposure is composed of two or more periods of noise exposure of different levels, their combined effect should be considered, rather than the individual effect of each. If the sum of the following fractions: C(1)/T(1) + C(2)/T(2) C(n)/T(n) exceeds unity, then, the mixed exposure should be considered to exceed the limit value. Cn indicates the total time of exposure at a specified noise level, and Tn indicates the total time of exposure permitted at that level. Exposure to impulsive or impact noise should not exceed 140 dB peak sound pressure level.

Continuous exposure according to OSHA entails not only a consistent noise at a stable dB level but also periodic noise that occurs at one second or fewer intervals. Additionally, seen in the footnote of the above table, OSHA requires consideration for exposure to noise of different dB levels for different periods of time. In such scenarios, hearing protection must be provided based upon the culminated exposure. For a typical patient being rescued by civilian helicopter rescue teams, exposure times can vary from a half-hour to several hours. In the case of military helicopter rescues and retrievals, the rescue times can also vary, but the maximum reported by the sponsors was on average one to two hours. By OSHA's standard, hearing protection must be provided for sound exposure of one or two hours at greater than or equal to 105 or 100 dB respectively. As previously mentioned, the maximum decibel level a helicopter's noise can reach is approximately 130 dB. Therefore, hearing protection is necessary for the flight crew and medical staff both inside the cabin and outside the immediate helicopter. However, this is not true for patients and victims who, for most search and rescue missions, are kept onboard without hearing protection due to having

injuries that are potentially much more severe and may be exposed to the high noise levels for only a portion of the total mission.

4) Customer Requirements and Engineering Specifications

The product's primary purpose is to provide hearing protection to its user, and the best way to measure its success is by the number of dB it reduces the incident sound pressure level on the user's ear by. Based on existing hearing protective technology (especially the current headphones used by the Air Force crew) and safe sound level measures provided by OSHA, the target reduction level for this product is 10 dB. Headphones currently used by crew members, such as those made by David and Clarke, and advanced ANC headphones such as those made by Bose, can reach sound reduction levels up to around 20 dB [3]. A goal of 10 dB is believed to be realistic given this existing technology and will noticeably improve the patient's experience.

For this product to be useful for most patients, it should be designed to meet as many worst-case scenario requirements as possible. This could entail the patient having a severe head or neck injury that may interfere with any devices placed on or around their head, not being in a conscious state, or needing active medical assistance throughout the rescue mission. Customer requirements to meet these needs include the product being able to provide protection without coming in direct contact with potential injuries or open wounds and should not envelop the patient so tightly that they feel claustrophobic. Additionally, the product being easily removable, having sections that can be opened, or not obstructing the patient would prevent this as well. The product needs to be applicable to the primary age group of rescued survivors, namely young to fully grown adults, which relates to the sizing and fit of the design.

Ideally, the product should provide little to no interference such that the medics can perform as many necessary tasks as possible with it employed. Additionally, having it be compatible with common medical devices such as heart rate monitors, respirators, or other measurement tools has been determined to be a stretch goal (not necessary for success but will be strongly considered), because this would be beneficial to the medics.

Finally, the Air Force and helicopter crew are also important stakeholders as they will be purchasing and implementing the device. Requirements to meet the needs of the Air Force include creating a product that is safe, complies with previously discussed standards, and is low in cost. An important requirement for the crew is ensuring that any light emission that the product creates is night vision goggle (NVG) compatible, as rescue missions carried out at night require NVG use and being exposed to a regular light source while wearing them can cause severe vision impairment. The device should also be durable and able to perform for at least the duration of a single rescue mission, although ideally, it could last several missions before needing to be replaced.

MALINE is especially aware of how the device will affect the patient and medical crew's experience as they are being transported. Frequent communication is conducted to ensure design modifications meet the goals and requirements of the Air Force squadron members who are the primary points of contact, but communication with people who have been rescued and have experience being transported via a Stokes litter is limited. Additionally, contacting members of the medical team is challenging as well. Thus, the team is actively trying to anticipate the needs and desires of these stakeholders (the patients and medical crew) who are not frequently providing feedback throughout the design process.

There are several design challenges that will need to be solved for this product to be a success. As discussed, creating a device that is easy for crew members/medics to implement while not negatively impacting the patient's experience is critical. This means ensuring hearing protection is provided while still allowing the patient to breathe regularly and be aware enough of their surroundings that they are not under unnecessary stress. More technically, a significant limitation is that this device must be operable while in flight on the helicopters during the rescue mission. Additionally, the device must not interfere with any of the frequencies used for communication within the helicopter or between the helicopter and external sources.

Table 2. Engineering Specifications Sheet for Sound Shell

Specification for: *Sound Shell*		
Requirement (leads to engineering specifications)	**Resp.**	**Source**
Create an environment where the noise of the helicopter and its surroundings are attenuated, giving the patient onboard a quieter trip.	MALINE	NSIN
Geometry Maximum Device Volume: 80" W x 84" L x 48'' H Maximum Device Weight: 100 lb.	MALINE	Captain Cooper [6] Captain Cooper [6]
Performance Minimum nose level reduction of 15 dB Night vision goggles (NVG) compatible Assembly time <= 30 min Implementation time <= 1 min	MALINE MALINE	Gentex Corporation Captain Cooper [6] Luis Santana, Michael St. John & Jim Webster Luis Santana, Michael St. John & Jim Webster
Energy Maximum power usage: 2.3 kW (20 A x 115 V)	MALINE	Captain Cooper [6]

Safety	MALINE	
Maximum localized temperature: -40ºF to 140ºF	MALINE	Richmond
Nonflammable and Noncarcinogenic		PHSMA
Radiation emission of <0.002 rem/hour		NRC, 10 CFR
		Subpart D
Either no light source or one that is compatible with NVGs		Captain Cooper [6]
Edges with radii of curvature >0.005 inches (No sharp Edges)		EHS Today
Maintenance	MALINE	IEC
Water Resistance (IP 68)		IEC
Dust Resistance (IP 68)		NEMA
Corrosion Resistance (NEMA 6P)		
Material	MALINE	PHSMA
Nonflammable and Noncarcinogenic		Topworks Plasticmold
Average surface roughness with Ra < 1 micrometer		
Ergonomics	MALINE	Captain Cooper [6]
Total weight: < 60 lbs.		EHS Today
All exposed edges: sanded and deburred (smooth to touch)		
Cost	MALINE	GT ME Dept.
BOM total expenditure: $4000 or less		NSIN

There are also several specifications the product must comply with based on information provided by the sponsoring members of the Air Force and additional rescue teams that have been contacted. A list of these specifications taken from the generated House of Quality is shown below in the specification sheet in Table 2. These include fitting within dimensions of 80" (width) x 84" (length) x 48" (height) and weighing less than 60 lbs. Values for the dimensions and weight were determined based on specifications for the interior of the helicopter and the weight that can be safely transported by the rescue team. Additionally, the total cost to develop the prototype must be under 4000 dollars based on the budget available to the team provided by the National Security Innovation Network (NSIN) and Georgia Tech Mechanical Engineering department. Based on existing technology and OSHA standards for safe sound levels, this product should achieve a minimum of 10 dB reduction in the SPL experienced by the patient. All materials used are required to be non-toxic and non-carcinogenic and should have a waterproof and dust resistance rating of NEMA 4X or IP65/66 [7]. The device should also emit 0 lumens of non NVG compatible light. It should be able to withstand temperatures resistant between –40 and 140 degrees Fahrenheit, and ideally will be fireproof and sand resistant to ensure it is safe to use in all conditions that the rescue team may be in.

5) Market Research

The market for this product is currently limited to the US Air Force, and more specifically the 41st Rescue Squadron. The current goal is to develop a model that is roughly $1000 for sale price as it will only include passive noise control (PNC). Note that these numbers are rough estimates. Once testing of specific acoustical materials is completed, and manufacturing costs are considered, then more accurate target prices will be finalized. The go-to-market strategy is to introduce the product to the Moody Air Force base and NSIN. If this goes well, then ideally the design would be implemented strictly into the helicopters in the military and air force. Once the device can prove its effectiveness in this field, the potential for use in other applications such as civilian rescue operations and even potentially at mining or construction sites can be evaluated.

The design team initially had numerous ideas about a potential design to pursue. However, market research, alongside the constraints of the problem, helped to impact the final design choice. For example, market research led to the team discovering the difficulties of active noise control (ANC) and how that would need to be a stretch goal for the product design. In addition, many solutions were not feasible for a helicopter due to either the inherent designs of the products or its expense. After conducting market research and discussing what is feasible, passive noise control (PNC) was selected to be the foundation for the product design.

6) Proposed Design Solution

The nature of a patient's injuries can be unknown and vary in severity; as such, a device that contacts the patient's head in certain locations is not guaranteed to work in every scenario the rescue team will face. Creating a solution that instead provides an enclosure that supports the patient's head on the Stokes litter without contacting other parts of their head or neck is a far more versatile option that will ideally accommodate greater numbers of rescued survivors. The Sound Shell, a helmet-like device that is fixed to the Stokes litter, accomplishes this. The design for the Sound Shell is based around two primary pieces: a top shell and a bottom base, which are to be joined together with multi-lock hinges. The full design is displayed in Figure 3.

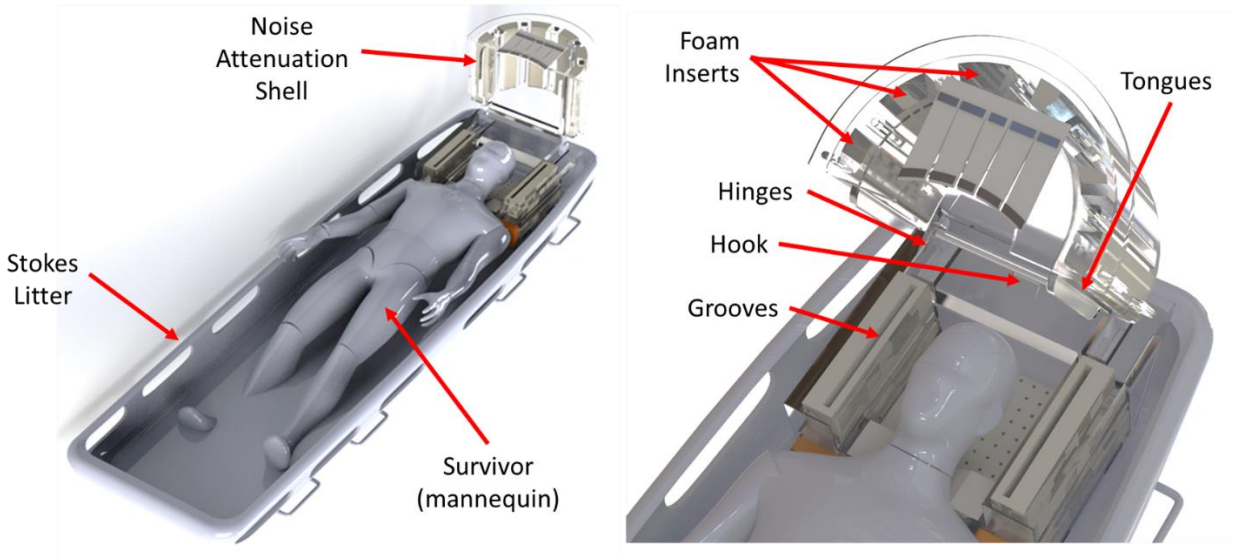


Figure 3: CAD Renderings of the device placed onto a modelled Stokes litter with patient

The bottom base piece is what is fixed to the Stokes litter where the patient's head would rest. The bottom piece features side walls that match the cross-sectional shape of the Stokes litter and includes channels to fit straps that anchor the device to the Stokes litter. The bottom piece possesses foam where the head and neck would rest. This foam, namely that near the neck region, serves both to cradle the person's neck and to reduce the size of the neck opening, thereby further dampening sound. In the region near the back of the head, there is foam present to provide comfort as well as some noise attenuation. This foam possesses drain holes, which lead to drain holes on the bottom of the base, allowing for drainage of liquid in cases such as a seawater rescue, thereby preventing drowning of the victim. The bottom section also includes an adjustable strap for the person's head. This strap is to be used to tie down the patient's forehead to keep their head in place within the shell to eliminate head jerking movements and reduce further injuries caused by such spinal stress. It is attached to the device through snap-fit buttons and loops about embedded bars on the inside of the base as well as Velcro. Another feature of the bottom piece is the presence of shoulder pads to provide support for the patient's shoulders, preventing them from being in contact with hard plastic. Lastly, a final feature that the base has is a hook that enables the Sound Shell to be attached to the cargo rings on the helicopter cabin walls (Figure 3). This allows for easy storage and keeps it out of the way until needed. Labeled pictures of the bottom base piece with visuals of the additional features are shown in Figures 4 and 5.

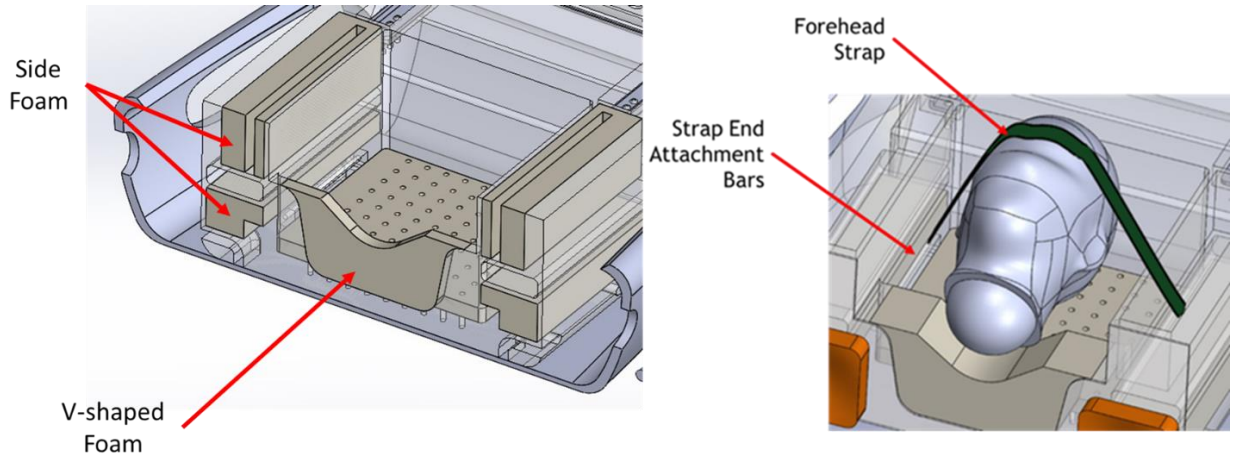


Figure 4. CAD Renderings focused on the device's bottom piece. Left is a sectioned view and right depicts the strap intended to go over the patient's forehead

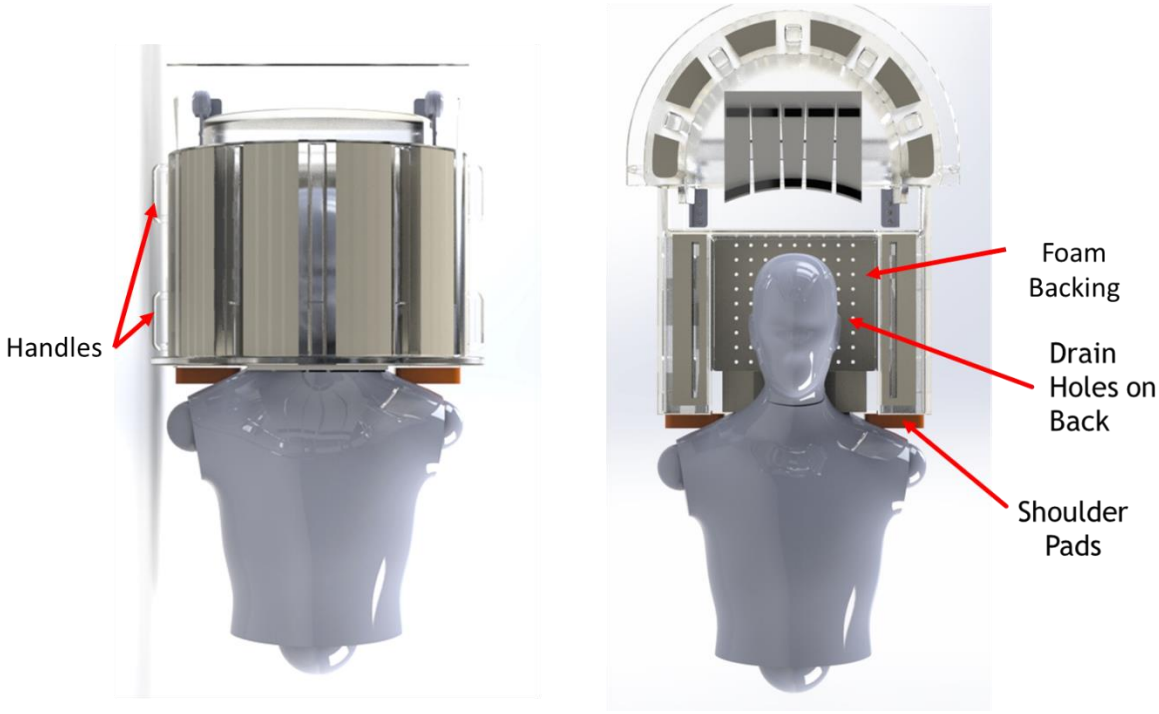


Figure 5. CAD Renderings of the device from a top-down view with the shell closed and opened, displaying the drainage holes and shoulder pads

The upper piece is a shell with a semi-circular arc cross-section that can be opened and closed, as it is fixed to the bottom piece on hinges located away from the crown of the patient's head. The shell opens away from the chin, akin to a car hood, and can be locked in place in several positions from 0 to 90 degrees as the hinges are multi-lock. The hinges are screwed in place into the bottom piece. Since the hinges are multi-lock hinges, they allow for the shell to be locked in an open or closed position without any user input

(i.e., someone holding the shell in place). The semicircular arc layers of the shell will most likely be made of silicone which is a plastic that will provide the supporting structure for the entire shell. Between the inner and outer layers of the shell are alternating inserts of acoustic foam and plastic. On the front face of the shell is a lid/cover that can be pulled off or pushed on to expose these inserts in the case where the acoustic foam needs to be replaced or removed. The foam will be wrapped in nylon or vinyl to allow it to resist water exposure, to be easily moved within the channels, and to be cleaned if needed. This foam, which will be Bass Buster (recycled acoustic bonded cotton fiber) foam, serves to dissipate as much noise as possible. The plastic located in between the foam inserts primarily acts to provide regions of transparency, permitting the crew to see the patient. Flaps on the front face of the shell serve to cover the neck opening providing further acoustic protection. These flaps have Velcro on them which can be brought together to reduce their length. This allows for the accommodation of various neck sizes and patient comfort. There are sets of two handles located on either side of the shell which can be used to easily lift the shell from the side. The shell's bottom sides possess tongues that slot into grooves dug into the side portions of the bottom piece of the device, eliminating gaps, securing the shell into the base, and permitting straightforward removal. An image of the upper shell with its labeled features can be seen in Figure 6.

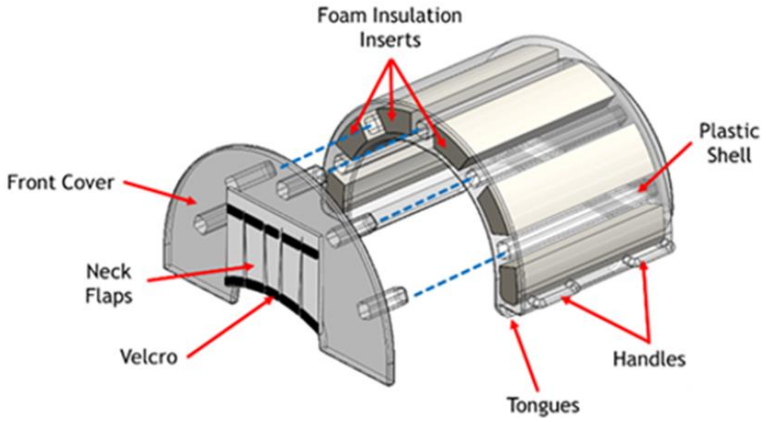


Figure 6. CAD Rendering of the top shell of the device, depicting removable front lid, foam inserts, plastic handles, and neck flaps

The overall dimensions of the Sound Shell are 19.05 in. x 18 in. x 19.26 in. as shown in Figure 7, and it is expected to weigh 55.2 pounds. The shell swings from 0 to 90 degrees, and at its maximum angle the height is 26.25 inches. Although the Sound Shell can be placed anywhere within the Stokes Litter, it will typically be placed near the ends. The Sound Shell is designed such that the 0-to-90-degree motion will not interfere with the Stokes Litter and can be placed in contact with the inner edge of a Stokes Litter rail.

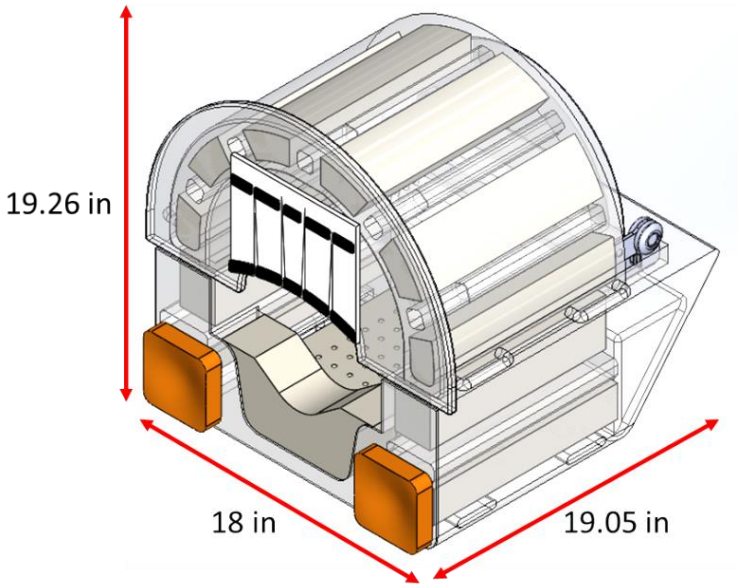


Figure 7. CAD Rendering of the current Sound Shell design with annotated dimensions

Lastly, the device is quick to assemble and clean. The device can be placed into the Stokes litter and strapped on during flight. This procedure will not require much time, which is critical due to the level of urgency typical of rescue missions. During retrieval of the patient, as the shell is already open on-site, there would be no impact on the placement of the patient onto the Stokes litter with the Sound Shell. Regarding cleaning, most of the device can be easily wiped and disinfected as significant portions of it are composed of plastic. There are certain crevasses that may be difficult to clean, particularly in the side portions of the bottom base, but they are still able to be wiped down and sprayed with disinfectant. Methods of cleaning the foam will vary. Some pieces are placed within parts of the device where it will not require cleaning, such as the foam within the shell or bottom base that is surrounded by plastic. Other pieces may be covered with vinyl or nylon, allowing it to be wiped down, while others are fully replaceable, such as the neck and backing foam in the bottom base. Vinyl and nylon-covered foam make those pieces much easier to clean and keep them resistant to liquids, especially seawater. Uncovered foam, like the neck and head support pieces, are expected to be replaced after every use due to hazardous exposure to biological material like blood or pollutants.

7) Industrial Design

Ensuring that the product is marketable by targeting a specific audience is not a concern given that the client is the Air Force, and its intended use is in medical emergencies. However, several considerations were made during the design process to ensure the device is safe to handle and visually appealing. One of the most significant of these design considerations was the selection for the device to open towards the top

of the patient's head, as opposed to opening to the side. This was decided on since there is very limited space in the helicopter cabin, and it is likely that medics will be to the sides of the patient. However, the patient's head will be at the top of the Stokes litter which will be up against a cabin door. Having the Sound Shell open upwards instead of to the side will allow for easy access to the patient for the medics on either side of the patient. Adjusting the design, however, to include a double hinge that would allow for opening both upwards and to the side would be a straightforward modification and is strongly being considered.

Additionally, considerations regarding the design and placement of the handles on the Sound Shell were taken. Having the handles be hollow cutouts into the shell was considered, but ultimately rejected, as these were believed to be more difficult to grab from various angles. It was decided that the handles should be on both sides of the shell to ensure that the device can be opened from several positions relative to the patient. The handles would be thin extrusions from the shell, as these would be easy to grab and maneuver.

Less obvious design considerations that were still made with the intent of making the Sound Shell user-friendly include several fillets to avoid exposed sharp edges and hinges hidden from being exposed to prevent rust or materials catching on them. Additionally, means of holding the different pieces together were designed to ensure that the device would be as easy to assemble and disassemble as possible. An example of this is the upper and lower portions of the shell being held together in the closed position simply with a snug fit, instead of using more time-consuming methods such as buckles or straps.

As for aesthetics, the design will likely feature a team logo, the Air Force logo, and the search and rescue logo of the Air Force. These aesthetic components are not necessary but provide an additional layer of appeal for the client. If these features are to be included, they would be relatively low in priority, being done after the main design of the product is completed.

8) Testing

Testing allows for the validation of the frequencies at which the helicopter will be operating and causing noise. Based on the Black Hawk 1996 Helicopter Assessment Paper, the highest sound pressure levels in dB in the helicopter interior occur at low frequencies [2]. This data was validated when MALINE went to the Moody Air Force base and was able to use a sound meter (Figure 13) to collect data from the HH-60W "Whiskey" Blackhawk Helicopter, which had its two engines on and was hovering. A graph of the data collected is shown in Figure 14. The graph depicts the frequency in Hertz on the x-axis and the sound pressure levels in dBA on the y-axis. It is important to note that the A-weighting characteristic of dBA measurements weighs higher frequencies stronger than lower frequencies, whereas C-weighting (for dBC measures) weighs low and high frequencies equally. Although helicopters typically have a sound level of 130 dB, the sound peaks on the graph show a peak of roughly 84.9 dBA. This may be partially due to the dBA scale being used and the 150 feet distance between the sound meter and aircraft. If the dBC scale

had been utilized, peaks at lower frequencies would have likely been higher. Due to this discrepancy, MALINE will be sure to acquire dBC data for the testing of potential materials. However, the primary purpose of the data acquired at Moody was to demonstrate that the maximum sound pressure levels occur in the lower frequency range of 1 to 1000 Hz with peaks around 500 Hz. This means that the materials that are being looked to construct the Sound Shell are usable and that the data found in Black Hawk 1996 Helicopter Assessment Paper are valid [2].



Figure 13. The sound meter used to collect the data of the helicopter frequency and decibel levels

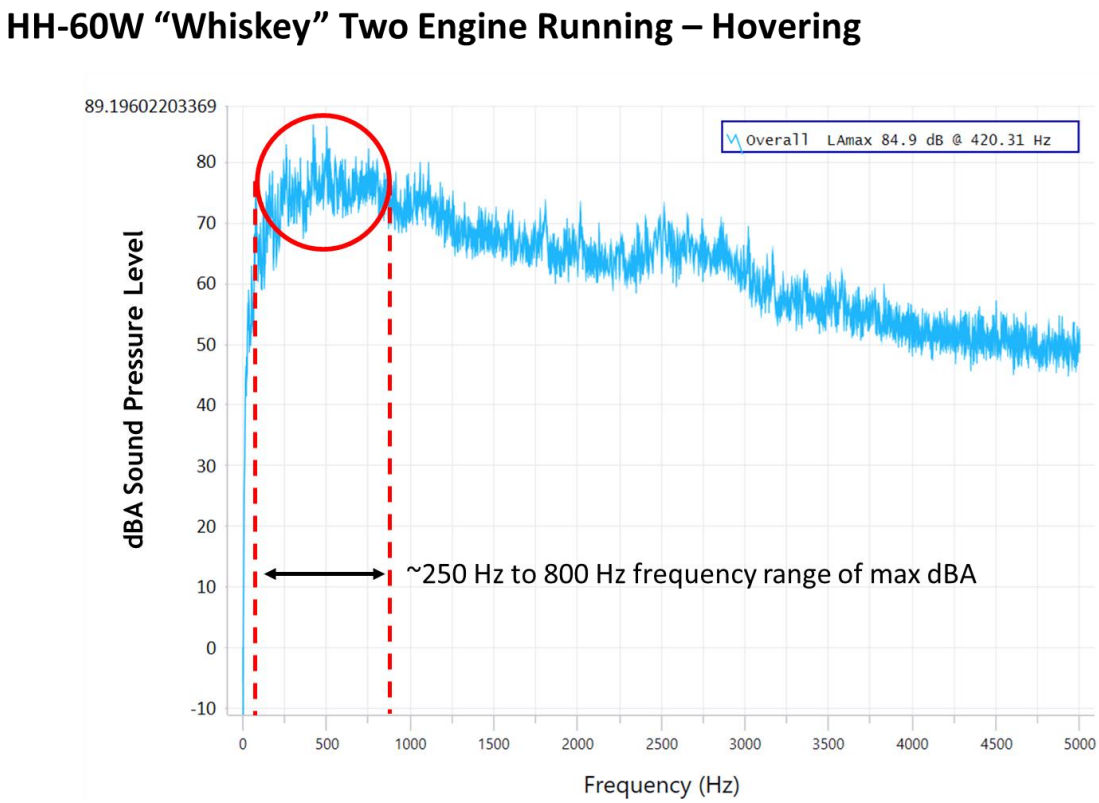


Figure 14. Graph of frequency versus dBA sound pressure level for two "Whiskey" engines running for the validation that higher dBA occurs at lower frequency range

To make an informed decision on what acoustic foam would be used in the Sound Shell, several rounds of acoustic testing were conducted. This process also helped inform whether the final design for the Sound Shell could be expected to achieve the 10-dB reduction in noise level that was set as a goal. The materials tested during this process included a semi-rigid acoustic paneling, a thin polyurethane acoustic foam, a Bass Buster foam (sold by Acoustical Surfaces), and the paneling and thin acoustic foam in combination. It should be noted that the Bass Buster is a four-inch-thick cotton fiber foam made specifically to attenuate low frequency noise, which as mentioned is pertinent for this design problem. The experimental set up for testing utilized a Fender bass amplifier connected to a laptop which was positioned [~6 ft] away from a Larson Davis sound meter. The bass amplifier was used to play noise at a fixed volume setting at frequencies ranging from 200 to 5000 Hz. Frequencies below 200 Hz were not compatible with the bass amplifier used. Baseline measurements were taken with nothing covering the sound meter, and measurements were also taken with solely a cardboard box rig covering the sound meter. This cardboard box was then used as a frame to attach the previously mentioned acoustic materials via Velcro. 5 rounds of testing were completed for each material, and results were averaged. All measurements were taken at the dBC sound pressure level to ensure low frequency signals were weighted similarly to high frequency ones.

It was observed that across the tested frequency ranges, the Bass Buster most consistently had the largest reduction in decibel level from the baseline measurement with nothing covering the sound meter. The results are shown in Table 4 and Figure 15a below. Figure 15b displays the reductions from baseline for all the materials tested. Following the analysis conducted in Table 4, the Bass Buster material was selected as the acoustic foam to be utilized in the Sound Shell. It should be noted that at 200 Hz, higher sound pressure levels were observed for all acoustic materials than for the baseline. This is potentially due to human error, as it was found that the head of the sound meter may have been in contact with the side of the rig for these measurements. Another potential reason is that 200 Hz was the resonant frequency of the testing floor. Since the bottom of the rig was left open, exposing the sound meter directly to the floor, the floor vibrating excessively at its resonant frequency may have led to the significantly higher values recorded at 200 Hz. This anomaly of having a dB reading higher than that of the baseline observed at 200 Hz is not believed to be characteristic of the Bass Buster since it was observed for all the acoustic materials (Figure 15b), and instead believed to be due to some other acoustic phenomena such as what was previously mentioned. A way to remedy this is to use rigs that have all sides covered in insulation or use a testing speaker that is not in contact with any solid surfaces.

Due to the Bass Buster having superior performance compared to the acoustic paneling and thin acoustic foam, it was selected to be used as the foam in the Sound Shell design. Other factors such as price and manufacturability also went into this decision and are discussed in the following section for both the Bass Buster foam as well as the other materials used in the final design.

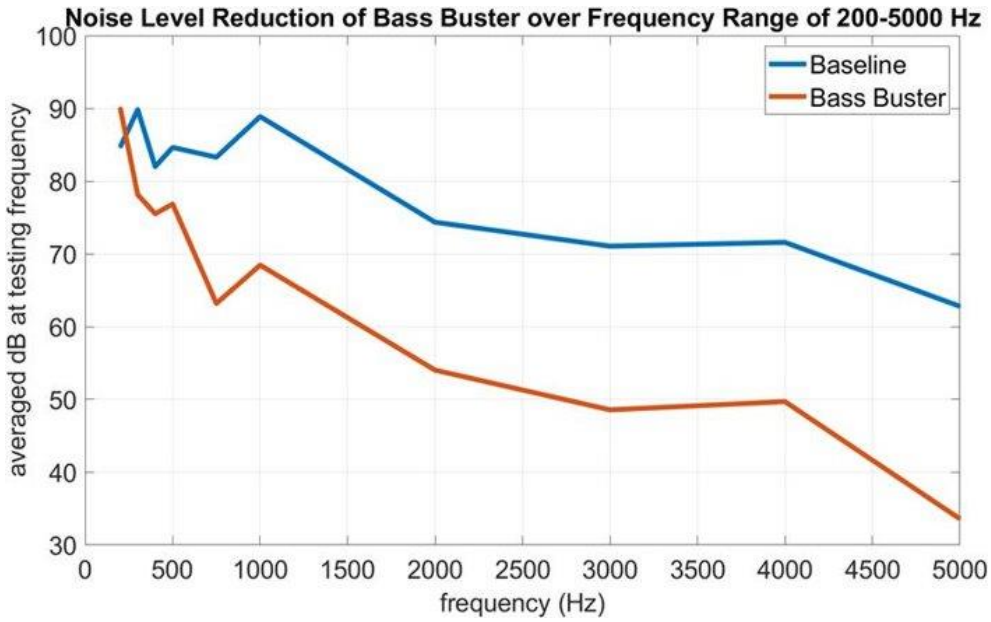


Figure 15a. Collected Noise Level Data for Bass Buster Foam and Baseline (with nothing covering sound meter).

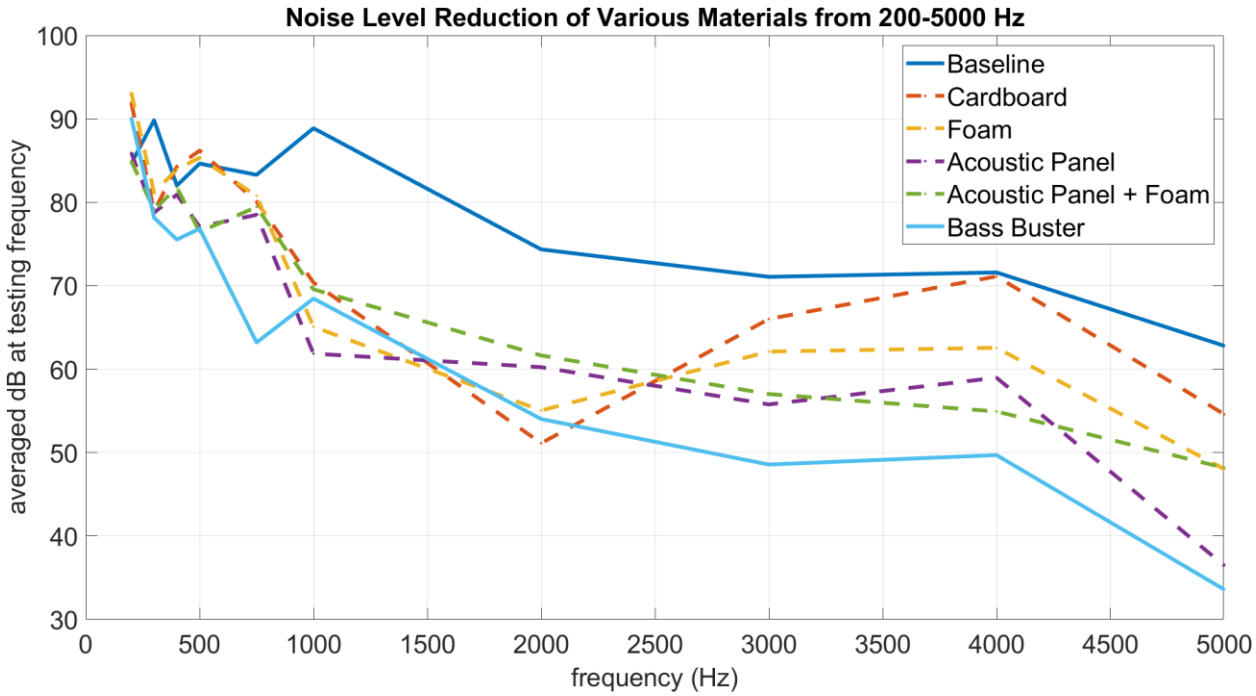


Figure 15b. Collected Noise Level Data for All Tested Acoustic Materials and Baseline (with nothing covering the sound meter).

Table 3. Reduction in Decibel Level from Baseline Measurements of Various Acoustic Materials.

	Difference in Sound Pressure Level from Baseline (Baseline – Material) [dB]				
Frequency [Hz]	**Cardboard**	**Acoustic Foam**	**Acoustic Panel**	**Acoustic Panel + Foam**	**Bass Buster**
200	**-7.3**	**-8.4**	**-1.3**	**-0.3**	**-5.4**
300	**10.9**	**8.7**	**11.1**	**10.7**	**11.7**
400	**-2.3**	**-2.0**	**1.1**	**0.2**	**6.5**
500	**-1.6**	**-0.7**	**7.6**	**8.2**	**7.8**
750	**3.2**	**2.5**	**4.8**	**3.9**	**20.1**
1000	**18.5**	**23.8**	**27.0**	**19.3**	**20.4**
2000	**23.2**	**19.3**	**14.1**	**12.7**	**20.3**
3000	**5.1**	**9.0**	**15.3**	**14.1**	**22.5**
4000	**0.5**	**9.0**	**12.6**	**16.7**	**21.9**
5000	**8.2**	**14.8**	**26.3**	**14.6**	**29.2**

There are several issues with the test setup used that should be acknowledged. The first is that the room testing was conducted in was a standard conference room, which lacked both the geometry and material properties of the interior of a helicopter. Due to this, the acoustic behavior of generated noise signals was most likely different. The way that sound would get reflected in a smaller space that has metallic walls was not able to be emulated, and sound was not generated from multiple different locations as it would be in a helicopter. Additionally, the material for the structure that all acoustic material options were fixed to was cardboard when it would be a plastic, such a silicone, on the Sound Shell. Also relating to the structural set up for testing, a cardboard box was utilized and therefore did not provide the same half cylinder geometry that would be the case in the Sound Shell. All these limitations were strongly considered before testing was started, and still should be considered when analyzing the results. However, the goal of this testing was to obtain a difference in measurements between a baseline setup, with no foam/paneling material present, and a test set up with foam/paneling present. Since all the limitations related to the room, cardboard, and geometry were the case for both the baseline and test measurements, the decibel reductions achieved by the acoustic materials is believed to be very close to what they would be if the same materials were being utilized in the Sound Shell within a helicopter.

9) Material Selection

One of the main challenges in the production of this design is finding and selecting materials that not only achieve the desired sound reduction but are also lightweight, non-toxic, and affordable. As mentioned in the previous section on the details of the design, the device will include several layers of plastic and foam to utilize both individual acoustic dampening properties as well as the benefits of acoustic impedance differences between each layer to further provide sound reduction. Several plastics were evaluated for potential use for the base and shell of the device. These include Silicone, PET, PMMA, and more, shown in Table 4 below. Ultimately, silicone was selected due to it having superior sound dampening properties compared to alternative options. Silicone has good sound dampening properties, specifically when concerning vibratory motion and forces. Since the helicopter vibrates when in use, the silicone can dampen those vibrations, allowing for the patient to be more comfortable. Not only this, but the sound waves that the patient is exposed to will also be attenuated by the silicone. When cured, silicone is also transparent and will therefore allow for visibility between the patient and medic. Finally, silicone can be processed to be rigid as it would be required to be in the current Sound Shell design. However, utilizing silicone in its flexible form could also allow for the device to be flexible and collapsible, with the structural support provided by the pieces of foam once they are inserted. If desired, this design modification could allow for the Sound Shell to take up less space when it is not being used.

As mentioned, the inserts of foam inside the shell will potentially be made of the Bass Buster foam that it is designed specifically to reduce low-frequency noise levels, which is especially applicable in and around a helicopter. An additional benefit of the Bass Buster foam is that it is made of cotton fibers and includes no glass fibers, making it safer than some alternatives for human exposure. Further details on the Bass Buster foam are shown in Table 4 [8]. All foam included in the design will be wrapped in vinyl, like a life vest, to ensure the device is easy to clean and can be exposed to saltwater. Flexible flaps made of polyurethane or polystyrene will be used above the patient's neck to provide additional acoustic protection without tightly enclosing the neck area. Table 4 below is the completed table showing materials, their potential use within the design, and relevant properties. Data on alternative materials and environmental considerations such as water consumption and carbon dioxide emission for production and processing was also collected and is shown in Tables 1A and 2A of the Appendix respectively. Due to relatively small differences in the data for this found and the low number of devices that are expected to be needed, this data did not have a large impact on material decisions made. However, if this is significant to the client, alternative material selections can be made.

Table 4. Specifications for Plastic Material Options.

Material	Use in device	Density [lb./in^3]	Cost [$/lb.]	Acoustic Impedance [MRayls] [9]	Service Temperature Range [°F]
Silicone [11]	Shell	0.0253-0.137	2.8-3.2 [10]	~2.0	-60 - 480
PET [11]	Shell	0.0466 - 0.0502	0.472-0.64	X	-70 - 140
PVC [11]	Shell	0.0466-0.0527	0.748-0.871	3.27	0 - 140

Table 5. Final Selected Design Materials with intended use and relevant material and acoustic properties. An entry of "X" denotes that the property was not able to be found.

Material	Use in device	Density [lb./in^3]	Cost [$/lb.]	Acoustic Impedance [MRayls] [9]	Service Temperature Range [°F]
Silicone [11]	Shell	0.0253-0.137	2.8-3.2 [10]	~2.0	-60 - 480
Polyurethane Foam [12]	Foam in neck, head, and shoulder support	0.000289-0.0504	0.544 - 0.744	1.80	-300 - 350
Bass Buster (recycled bonded acoustical cotton fiber) [8]	Foam in shell	0.000326	3.67	X	X
Vinyl (ethylene-vinyl-acetate) [11]	Foam wrapping	0.0341-0.0345	0.626-0.64	1.69 (EVA) 2.96 (rigid)	-190 - 120

Table 6a. Bill of Materials for Full Assembly

Item Number	Vendor	Part Number	Description	Quantity	Price ($)
1	N/A	N/A	Shell Subassembly	1	452.90
2	N/A	N/A	Base	1	421.67
3	N/A	N/A	Back of Head Foam Support	1	0.74
4	N/A	N/A	Next to Ear Foam Support (2 total)	1	2.59
5	Sugatsune	HG-MA95BF-R	Right Angle Hinge	1	33.06
6	McMaster-Carr	93365A154	Tapered Heat-Set Inserts for Plastic	10	2.41
7	McMaster-Carr	92949A271	18-8 Stainless Steel Button Head Hex Drive Screw	10	1.89
8	N/A	N/A	Neck Foam	1	0.69
9	Sugatsune	HG-MA95BF-R	Left Angle Hinge	1	33.06
10	N/A	N/A	Bottom Piece Foam Inserts (2 total)	1	2.51
11	N/A	N/A	Front Cover of Bottom Piece	1	4.85
12	N/A	N/A	Shoulder Padding	1	31.00
13	N/A	N/A	Model of human (Not needed)	1	
					$987.37

Table 6b. Bill of Materials for Shell Subassembly

Item Number	Part Number	Description	Quantity	Price ($)
1	N/A	Shell	1	327.71
2	N/A	Shell Foam Inserts	1	5.73
3	N/A	Shell Front Cover	1	54.76
4	N/A	Front Flaps	1	64.70
				$452.90

Based on tables 6a and 6b, the material cost of the Sound Shell would be less than $1000, as mentioned previously. This estimate does not include the cost to manufacture the device, and potential methods of doing so is discussed in the following section. Material selections were made with reducing the overall cost in mind, in addition to the previously discussed reasons that work to achieve the engineering requirements of the device, and so it is not believed that the cost can be further reduced by selecting alternative materials. However, this cost can potentially be reduced in the future if testing of an accurate prototype proves that a 10-decibel reduction can be achieved even when using a smaller volume (ex. lower thickness) of the selected materials.

10) Manufacturing

There are several methods that can be utilized to manufacture the Sound Shell, all with their own benefits and downsides. Based on the complexity of the plastic portion of the design and the hollow interior channels that exist in several locations, MALINE suggests primarily relying on injection molding. The main benefits of injection molding are that it will be able to achieve detailed interior features, such as the hollow

channels for the foam, and it is the best option for mass production. As the number of parts created continues to go up, the marginal price of producing an additional unit is very low when compared to alternatives. Despite this, there is a very high cost for initial tooling requirements, which should be strongly considered if the number of units required is low.

An alternative to injection molding that may be better suited for producing less units is vacuum forming. The benefits of vacuum forming include that it is less complex and lower cost for a low volume than injection molding. However, vacuum forming will only be able to achieve simple designs and so more additional post processing steps will be required. Additionally, MALINE recommends using silicone for the plastic portion of the design as previously mentioned. However, silicone would not work with vacuum forming, and so an alternative such as PET or PVC would be required. These plastics would still achieve the transparency that silicone would while also still contributing to dampening sound. Thus, if vacuum forming is the best option for all reasons other than utilizing silicone, this should not be a significant barrier to utilizing vacuum forming over other most costly alternatives.

A final method for manufacturing is CNC-machining. This method is best for producing a very low number of units, as it is more time intensive and poor at fast, repetitive mass production. The primary benefit of CNC machining is that it would allow for all detailed internal features and would provide very good tolerance levels. Very precise tolerances are not of primary concern for the Sound Shell and so MALINE would recommend CNC machining the device only if a very low number of them are needed.

To assist with making a recommendation for manufacturing, MALINE reached out to obtain quotes for injection molding of the Sound Shell design. Specifically, the companies Protolabs and Xometry were contacted. From the quotes received, one piece of the Sound Shell is expected to cost $5,610 to manufacture. However, the unit cost is expected to significantly decrease as more Sound Shells are being manufactured.

11) Assembly

There are several steps that are required to assemble the various pieces of the Sound Shell. For example, the foam pieces that will be inserted in the upper shell and for the shoulder pads should first be wrapped in vinyl to ensure they are waterproof and easy to clean. Once all the foam has been inserted into the Sound Shell, the cover of the bottom piece is attached using acoustic caulk. Assuming this was not completed in initial manufacturing steps, holes must also be drilled, or threaded heat inserts should be added to previously created holes in the back of both the upper and lower pieces. The purpose of these holes is to attach the hinges to connect these two pieces, and so fasteners (included in the bill of materials mentioned previously) will need to be added to fix these hinges in place as well. The neck flaps must be fixed to the front cover of the upper pieces using the same acoustic caulk, and Velcro should be attached to the front of these flaps as shown in Figure 6. If the manufacturing engineer determines that creating the handles

separately from the shell piece will drastically reduce complexity, these pieces can be made individually and assembled on to the shell.

Regarding the maintenance of the parts of the Sound Shell, all the foam pieces made of the Bass Buster material do not need to be replaced unless there is considerable degradation over time due to environmental conditions, time, and general wear. These pieces can be replaced and swapped with new foam inserts if needed. The foam supporting the neck and the back of the head as well as the head straps will need to be replaced after every mission to maintain cleanliness. The vinyl coverings on the neck flaps and shoulder pads can be sanitized after each use.

12) Future Considerations

The current Sound Shell design provides a strong foundation for a device that provides moderate sound reduction, is easy to implement, and is cost-effective. However, given additional time and a larger available budget, there are several features that could be added to improve the effectiveness of the device as well as the customer's experience. The first of these is the addition of active noise cancellation (ANC) technology. Active noise cancellation utilizes a microphone, speaker, and digital signal processor to pick up sound signals and generate inverse signals that destructively interfere with the source, thereby reducing the signal. ANC was not pursued in the current Sound Shell design due to MALINE's limited electrical engineering experience, however, the design would allow for easy implementation of ANC if this were desired. This is because the Sound Shell is a relatively small, enclosed system and thus encapsulates the head and ears, making ANC easier to use (it would be much harder in a larger enclosed system, such as the cabin itself). Microphones could be set up at certain locations, such as where the most sound is being let into the helmet or adjacent to where the patient's ears are, and speakers could be placed adjacent to or just below the patient's ears within the helmet. The primary challenge with implementing ANC would be creating a signal processing system that is able to adjust to reflections inside of the helmet itself, as well as considering that the signals generated by the speaker that intends to cancel the source signal may interfere with the source signal that the microphone picks up. An additional challenge would be ensuring the ANC system, which is primarily electrical components, could be exposed to water as the Stokes litter may be used in water-based rescue missions.

The Sound Shell could also be improved by modifying it to be able to be utilized without being externally supported, such as how it currently is with the Stokes Litter. This would make it so that patients who are not being transported via the Stokes litter could also use it. Another added benefit would be that the medical team could transport the Stokes litter without the device attached and decide whether to place it on the patient's head before placing them on the Stokes litter based on the situation they observe. A key challenge with adding this ability is considering where the device would be supported, especially if the

patient has an injury at the shoulders or higher up their body. If the patient does not have injuries around their head, neck, or upper torso, a potential solution would be to add an additional modular attachment at the bottom of the device that allows it to rest on the patient's shoulders. Another challenge is its weight. The Sound Shell would need to be made much more lightweight. Lastly, the external support capability can be retained such that it can latch into the Stokes litter and onto the cabin wall of the helicopter if necessary. Changing where the device mounts (be it freestanding, on the Stokes litter, or on the cabin wall) would increase the scope necessary for the product and could potentially change the geometry needed for passive noise control (PNC).

Another design improvement that could be made for the future is modularity. The device could be made modular. By being modular, the Sound Shell would have multiple parts that could be disassembled and assembled. For instance, the best potential way to make the device modular would be to disconnect the shell from the base by allowing the hinges to slot in and out of a press-fit slot on the base. Thus, the shell could be pulled off completely and the base portion could be attached to Stokes litters in general with the shell placed on later. In addition, if the medics need quick access to the patient and do not care about their hearing being damaged (i.e., in life-or-death situations), the shell could be removed in its entirety and tossed to the side allowing for better visuals and the ability to treat the patient further.

Currently, the Sound Shell base and its shell are not hollow. A potential way to improve upon the design would be to make the base and the shell hollow such that the solid plastic only provides a "mold," encapsulating the foam. This would increase the acoustic impedance mismatches between the materials as the sound would now need to travel through an additional layer of air which would make it bounce and reflect within the base and shell instead of the cavity where the patient's head is located. To further expand upon this idea, the plastic "mold" could be gotten rid of in its entirety, instead using plastic to form a skeletal structure for the Sound Shell thus that sheets of plastic could be wrapped around the skeleton. This would achieve roughly the same goal as making the Sound Shell hollow.

Another design improvement would focus on the patient's comfortability within the shell. This would be achieved through research such as interviewing survivors that have been in Stokes Litters in helicopters. Their needs would be analyzed and reflected upon and features to address those needs would be added where possible. Not only this, but a microphone and speaker system could be added within the Sound Shell to allow the patient and medics to communicate over a radio frequency already more easily in use by the military. Overall, the patient's comfort is a priority that needs to be further addressed.

A final design improvement would be to make the device compatible with medical equipment such as heart rate monitors or respirators. If these devices were built into the Sound Shell or easily placed while it is in use, it would allow the medical crew to have more reliable means of monitoring the patient's well-

being while the Sound Shell is still in use. This would potentially reduce the frequency with which the device would be opened, or kept off the Stokes litter entirely, depending on the patient's condition.

Lastly, if the Sound Shell were to be used in other applications, it would be in other medical areas with high noise. For instance, a civilian helicopter, while more acoustically insulated than a military helicopter, still has a large amount of sound. In this case, the Sound Shell could be simplified further and adapted to that scenario. In addition, the Sound Shell could be modified to better protect the construction and mining communities, both physically and acoustically. This could potentially be done by making the plastic layers thicker and making the device freestanding (i.e., no need for an attachment point). Overall, the Sound Shell has a great deal of potential.

13) Conclusions, Future Work, Project Deliverables

Up to this point, MALINE has gained substantial information on the client's requirements for this device to be considered a success as well as on the specifications that the device must comply with. Additionally, extensive research has been conducted on topics related to the problem at hand, which include acoustics, helicopter specifications, fluid mechanics, and structural vibrations. From this, along with an ideation process, several iterations of the Sound Shell design were created. The Sound Shell provides a comfortable enclosure that reduces noise levels without coming in direct contact with the patient (other than supporting their head and neck) and allows for medics to continue to perform their roles. An upper shell that features both transparent plastic channels, as well as water-resistant acoustic foam, allows for both visibility and acoustic attenuation, and additional foam supports underneath the head and neck and above the forehead ensure patient comfort and safety. Ease of implementation both in terms of the ability to quickly open the upper shell when contact with the patient is needed and in terms of the ability to remove the device completely from the Stokes litter by undoing the attached straps also ensure that the Sound Shell will not interfere with the ever-changing situations that a rescue mission may involve.

Potential limitations of the Sound Shell include that it must be fixed to the Stokes litter before the patient is placed on it and that it must be opened (thereby exposing the patient to incident noise) to have direct access to the patient. However, it is important to note that the design provides a strong foundation with its use of passive noise control, and additional modifications such as the implementation of ANC or expansion to be used beyond just the Stokes litter can improve upon the design. Additionally, as testing of various materials and geometries is conducted, improvements that are viable within MALINE's time and monetary constraints will continuously be made to the Sound Shell before a final prototype is presented.

In terms of future work, there are several steps MALINE would take if given more time. The first would be to conduct additional testing. With this, geometries more like how the Sound Shell would look could be tested. Options for the plastic portion of the design, such as silicone or alternatives like PET or

PVC would also be tested. Work would also be done to create a full scale, accurate prototype that utilizes selected materials. This would not only allow the team to conduct more accurate testing to determine if the Sound Shell achieves at least a 10-decibel reduction but would allow the team to better understand if the device is as comfortable as its design intends. It would also help the team determine any limitations that may be present in the proposed assembly and manufacturing methods. Finally, significant efforts would be put into obtaining potential customer feedback. People who have been rescued via helicopter or carried in a Stokes litter would be contacted to better understand their needs and concerns. Additional features that would be added to the design would also primarily be related to attending to potential customers' comfort and safety.

14) Team Member Contributions

Mitul Pandya served as the primary informant regarding the search and rescue aspect of the project. This included communicating with multiple search and rescuers, policemen, and directors to better understand the search and rescue process. Included in this role was to ask the sponsor the relevant questions needed to bound the problem. Mitul also worked with Cole every Wednesday to help run the advising session with Dr. MacNair, spearheading any discussions and reviewing the progress the team made. Mitul also created and presented the first presentation with Eric to Dr. MacNair and other Capstone students, highlighting the progress the team had made. Regarding the report, Mitul worked on editing and reviewing the material as well as creating the evaluation matrix on the proposed solutions. Mitul also worked on the fabrication package for report 2. This involved designing and revising all the CAD parts in SolidWorks (3 iterations) and assembling them. This also included making the part drawings and bill of materials. Finally, Mitul worked on creating renders and animations for the second presentation. For report 2, Mitul helped work on the manufacturing section and editing and reviewing. Mitul went on the Moody Air Force base trip and used the sound meter to obtain measurements of the rescue helicopter startup and hovering stages. For report 3, Mitul creating the fabrication package and any additional renderings or animations. Additionally, Mitul created the fit test of the Sound Shell with the help of Eric.

Ayan Kadri served as the agenda agent for the team. As the agenda agent, Ayan would write agendas for each team meeting and keep the team on track during meetings. In addition, Ayan oversees keeping the Teams files organized so that everything is easily accessible. As for report 1, Ayan, alongside Ivan, worked to create function trees and morphological charts. Ayan contributed to helping Cole work on the market research and prior arts research. Ayan also worked extensively on the design. Ayan also worked with the other members of the team to conduct sponsor interviews and meet with professors to garner information. Ayan contributed extensively to the entirety of the design phase of the Sound Shell, helping the team to visualize and conceptualize different ideas. Ayan also conducted research based on the design

to make sure that the device had the potential to function. Ayan also went with the team to the Moody Air Force Base and helped to collect sound data from the helicopter. Ayan also helped to work on the presentation a little bit, helping Cole and Ivan practice. Ayan also did some preliminary research through GRANTA on materials. In addition, Ayan worked on the 2nd report and helped to flesh out the drafted sections, drafted the manufacturing section, and worked on revising and editing the rest of the report. Ayan worked extensively on the design portion, manufacturing, and future considerations portions. Ayan helped extensively with testing by guiding the data collection and helping to set up the testing rigs utilized. For report 3 and presentation 3, Ayan worked on both extensively, eventually presenting for presentation 3. Ayan was also in charge of getting the poster printed and putting together the binder for the Capstone Expo. Ayan also helped with the fit test. Overall, Ayan worked with his team to the best of his ability to maintain a high level of effort and productivity.

Lizzie Hoole served as the sponsor liaison and maintained contact/set up meetings with the various sponsors of this project each week. Additionally, Lizzie was the primary researcher for acoustic-related topics. Lizzie also helped with contacting external sources, planning for/conducting testing, and working with the team to come up with the selected design (and iterate to improve it). Lizzie drafted all sections of the second report except for the codes and standards and manufacturing sections, helped revise the report, helped with finding material properties, and helped with some of the simpler CAD package drawings. For the third report, Lizzie drafted the new sections added (testing, manufacturing, assembly) and updated the materials, future work, and conclusion sections. She also helped revise the remainder of the report and helped create the slides for the third presentation.

Ivan Li acted as team scribe, recording notes during meetings with team advisor Dr. MacNair, calls with Air Force sponsors, and collaboration sessions with the entire team. Outside of his assigned role, Ivan conducted research about codes and standards for noise protection and helicopter flight. Ivan contributed to the visual understanding of the design through sketches, drawings, and ideation. Ivan composed the parts of this report pertaining to standards, the overall final design, and overall editing. He, as with most of the rest of the team, went to the Moody Air Force Base to gain more insight into the equipment and aircraft used and collect data and recordings. He conducted testing and data analysis of the acoustic attenuation of materials in MATLAB and produced relevant figures. He assisted Mitul and Eric in prototype construction and assembly. Lastly, Ivan alongside the rest of the team presented the finalized current design, testing, and future work during the Capstone Expo.

Nicholas “Cole” Miller oversaw the market research and prior arts research alongside Ayan. This research was ongoing and constantly updated as the latest ideas were introduced. As a meeting manipulator, he worked with Mitul every Wednesday to create PowerPoints for MALINE’s advisor, Dr. David MacNair, and stimulated the discussion and oversight of the PowerPoints. As a general team member, Cole brought

forth ideas and questions for other sponsors, the advisor, and additional research professors. Cole was the main driver to the Moody Air Force base visit and offered his vehicle up for transportation. He oversaw the PowerPoint presentation alongside Ivan for presentation 2. He practiced the presentation with Ivan in front of the team multiple times before the actual presentation date. He met up for testing and brought the acoustic paneling and foam to and from meetings as needed. Cole also functioned as an editor to the second and final report. Cole worked with Ivan to perform the recent testing 5 times across the various frequencies and materials of interest.

Eric Son has been working as the budget organizer who continues to keep up with the equipment purchased for the design and equipment. He has been reaching out to the NSIN as well the school for financial support for the design and testing of this project. He has also been involved with reaching out to many faculty members and experts for acoustic research along with Ayan Kadri, and Eric gained more connections and contact with people to maximize access to various resources for design testing. He has also worked on the first presentation with Mitul Pandya. Along with most of the team, he went on the field trip to the Moody Air Force Base to collect data and gain information relevant to the project. He contributed to both the brainstorming for the general design as well as the test rig for acoustics testing. For the report, Eric was heavily involved in the graphs and figures as well as the overall editing with the feedback from the presentation taken under consideration. Eric also went to Home Depot with Mitul to get parts and glues for the prototypes. He helped attach the materials in rig testing and helped to evaluate the rigs as well. In addition, Eric helped extensively in assembling the fit-type prototype.

15) Acknowledgements

MALINE had help from several people throughout the duration of this project, without whom the team would not have been able to complete all of the work that they did. First and foremost, MALINE would like to acknowledge and thank Dr. David MacNair, who served as the team's faculty advisor and who was integral to the progress made on this project. Dr. MacNair provided feedback, advice, and direction and was critical to the team's success. Additionally, all the members of the sponsoring groups for this project were extremely helpful. Captains Cooper and Knapp of the U.S Air Force and Dr. Patrick Reynolds and Cassie Heyman-Schrum of the NSIN especially provided information and support that made a significant difference and was greatly appreciated. The team would also like to thank Luis Santana, Michael St. John, and Jim Webster for providing insight into the operations of helicopter search and rescue. Finally, all of the Georgia Tech faculty who assisted with the research portion of this project, and Dr. MacNair's other senior design teams who provided a great deal of feedback also were a significant help.

Appendix

Table 1A. Potential Design Materials with intended use and relevant material and acoustic properties. An entry of "X" denotes that the property was not able to be found.

Material	Use in device	Density [lb./in^3]	Cost [$/lb.]	Acoustic Impedance [MRayls] [9]	Service Temperature Range [°F]
Silicone [11]	Shell	0.0253-0.137	2.8-3.2 [10]	~2.0	-60 - 480
PET [11]	Shell	0.0466 - 0.0502	0.472-0.64	X	-70 - 140
PVC [11]	Shell	0.0466-0.0527	0.748-0.871	3.27	0 - 140
ABS [11]	Shell	0.0372-0.0383	0.78-0.912	2.31	-49 - 170
Acrylic Plexiglass/ PMMA [11]	Shell	0.0423-0.0434	0.816-0.871	3.26	-103 - 133
Mylar [11]	Shell	0.051	X	3.0	X
Epoxy [11]	Shell	0.065	X	X	X
Polyurethane Foam [12]	Foam in neck, head, and shoulder support	0.000289-0.0504	0.544 - 0.744	1.80	-300 - 350
Polystyrene Foam [11]	Foam in neck, head, and shoulder support	X	X	2.42	X
Polyethylene Foam [11]	Foam in neck, head, and shoulder support	X	X	1.79 (low density) 2.33 (high density)	X
Bass Buster (recycled bonded acoustical cotton fiber) [8]	Foam in shell	0.000326	3.67	X	X
Nylon (6/6) [11]	Foam wrapping	0.0405-0.0415	1.99-2.93	2.9	-110 - 260
Vinyl (ethylene-vinyl-acetate) [11]	Foam wrapping	0.0341-0.0345	0.626-0.64	1.69 (EVA) 2.96 (rigid)	-190 - 120

Table 2A. Environmental Impact Data for Potential Sound Shell Materials [11].

Material	Use in device	Water Usage (production) [in^3/lb.]	Carbon Footprint (production) (lb/lb)	Molding Carbon Footprint (processing) [lb./lb.]	Molding Energy (processing) [BTU/lb.]	Recyclable (energy required if applicable)
Silicone	Shell	5.26E3-1.58E4	6.2-6.83	1.13-1.24	6.06E3-6.66E3	No
PET	Shell	3.49E3-3.88E3	2.59-2.86	1.4-1.55	8.04E3-8.86E3	Yes (1.15E4-1.27E4 BTU/lb)
ABS	Shell	4.62E3 - 5.12E3	3.27-3.61	1.47-1.63	8.47E3-9.33E3	Yes (1.32E4-1.46E4 BTU/lb)
PMMA (Acrylic)	Shell	2E3-2.21E3	4.64-5.11	1.32-1.46	7.57E3-8.34E3	Yes (1.56E4-1.72E4 BTU/lb)
PVC	Shell	5.01E3-5.54E3	2.57-2.83	1.05-1.16	5.98E3-6.62E3	Yes (9.5E3-1.05E4 BTU/lb)
Polyurethane	Foam	2.59E3-2.85E3	3.05-3.37	1.27-1.4	6.84E3-7.52E3	No
Nylon	Foam wrapping	4.87E3-2.27E4	6.09-8	1.55-1.72	8.9E3-9.85E3	Yes (1.78E4-1.96E4 BTU/lb)
Vinyl	Foam wrapping	73.6-81.4	2.01-2.21	1.19-1.31	6.36E3-7.05E3	No

References

[1] "What is an NRC rating?," *Audimute*. [Online]. Available: https://www.audimute.com/nrc-ratings-noise-reduction-coefficient#:~:text=A%20Noise%20Reduction%20Coefficient%20is,those%20products%20can%20soak%20up. [Accessed: 16-Mar-2022].

[2] R. B. King, A. J. Saliba, D. C. Creed, and J. R. Brock, "Assessment of Noise Levels In and Around The Sikorsky S-70A-9 Black Hawk Helicopter," *DSTO Aeronautical and Maritime Research Laboratory*, vol. VIII, p. 46, 1996.

[3] "Fixed wing headsets (ENC)," *David Clark Company*. [Online]. Available: https://www.davidclarkcompany.com/aviation/fixed-wing-enc.php. [Accessed: 08-Feb-2022].

[4] "NoCry 6-in-1 Industrial Forestry Safety Helmet and Hearing Protection System; Face Protection Equipment with Two Protective Visors," Amazon. [Online]. Available: https://www.amazon.com/NoCry-Industrial-Forestry-Protection-Protective/dp/B07CX3QYHW/ref=asc_df_B07CX3QYHW/?tag=hyprod-20&linkCode=df0&hvadid=309781195607&hvpos=&hvnetw=g&hvrand=15165858248895914024&hvpone=&hvptwo=&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9060222&hvtargid=pla-569532043384&psc=1&tag=&ref=&adgrpid=61053849069&hvpone=&hvptwo=&hvadid=309781195607&hvpos=&hvnetw=g&hvrand=15165858248895914024&hvqmt=&hvdev=c&hvdvcmdl=&hvlocint=&hvlocphy=9060222&hvtargid=pla-569532043384. [Accessed: 09-Feb-2022].

[5] "Department of Labor United States department of Labor," 1910.95 - Occupational noise exposure. | Occupational Safety and Health Administration. [Online]. Available: https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.95. [Accessed: 09-Feb-2022].

[6] C. Cooper, Rescue Equipment Powerpoint, 01-Feb-2022.

"Acoustic properties of...," *Ultrasonic or ultrasound sound velocity and impedance | Signal Processing*. [Online]. Available: https://www.signal-processing.com/table.php. [Accessed: 16-Mar-2022].

[7] "NEMA ratings and what they mean," *Gore*. [Online]. Available: https://www.gore.com/resources/nema-ratings-and-what-they-mean#. [Accessed: 09-Feb-2022].

[8] "Bass Buster- Low Frequency Sound Absorber." [Online]. Available: https://www.asistorefront.com/p-223-bass-buster-low-frequency-sound-absorber.aspx. [Accessed: 16-Mar-2022].

[9] "Acoustic properties of...," *Ultrasonic or ultrasound sound velocity and impedance | Signal Processing*. [Online]. Available: https://www.signal-processing.com/table.php. [Accessed: 16-Mar-2022].

[10] F. C. I. 2022 (https://fci-nj.com), “Silicon vs silicone: Tariffs and naturally limited resources: Blog: First Continental International: Window to the Chemical World,” *First Continental International*, 06-May-2019. [Online]. Available: https://fci-nj.com/silicon-vs-silicone-tariffs-and-naturally-limited-resources/. [Accessed: 16-Mar-2022].

[11] CES EduPack Software, Granta Design Limited, Cambridge, UK, 2009. [Accessed: 16-Mar-2022].

[12] “Overview of Polyurethane Foam, Unreinforce,” *Overview of materials for thermoset polyurethane foam, unreinforced*. [Online]. Available: https://www.matweb.com/search/DataSheet.aspx?MatGUID=91d44cae736e4b36bcba94720654eeae. [Accessed: 16-Mar-2022].